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#include <stdio.h>

int main() {
    int n, m, i, j, k;
    printf("1WA24CS218\n");
    printf("Enter the number of processes: ");
    scanf("%d", &n);
    printf("Enter the number of resources: ");
    scanf("%d", &m);
    int allocation[n][m];
    int request[n][m];
    int available[m];
    int work[m];
    int finish[n];
    int safeSequence[n];
    printf("Enter the allocation matrix:\n");
    for(i = 0; i < n; i++) {
        printf("Process %d: ", i);
        for(j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }
    printf("Enter the request matrix:\n");
    for(i = 0; i < n; i++) {
        printf("Process %d: ", i);
        for(j = 0; j < m; j++) {
            scanf("%d", &request[i][j]);
        }
    }
    printf("Enter the available resources: ");
    for(i = 0; i < m; i++) {

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scanf("%d", &available[i]);
work[i] = available[i];
}
for(i = 0; i < n; i++) {
    int allZero = 1;
    for(j = 0; j < m; j++) {
        if(allocation[i][j] != 0) {
            allZero = 0;
            break;
        }
    }
    if(allZero)
        finish[i] = 1;
    else
        finish[i] = 0;
}
int count = 0;
while(count < n) {
    int found = 0;
    for(i = 0; i < n; i++) {
        if(finish[i] == 0) {
            int possible = 1;
            for(j = 0; j < m; j++) {
                if(request[i][j] > work[j]) {
                    possible = 0;
                    break;
                }
            }
            if(possible) {
                for(k = 0; k < m; k++) {

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        work[k] += allocation[i][k];
    }
    safeSequence[count] = i;
    count++;
    finish[i] = 1;
    found = 1;
}
}
}
if(found == 0)
    break;
}
int deadlock = 0;
for(i = 0; i < n; i++) {
    if(finish[i] == 0) {
        deadlock = 1;
        break;
    }
}
if(deadlock) {
    printf("\nSystem is in DEADLOCK state.\n");
    printf("Deadlocked processes are: ");
    for(i = 0; i < n; i++) {
        if(finish[i] == 0)
            printf("P%d ", i);
    }
    printf("\n");
}
else {
    printf("\nSystem is in safe state.\n");

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printf("Safe Sequence is: ");
for(i = 0; i < n; i++) {
    printf("P%d ", safeSequence[i]);
}
printf("\n");
}
return 0;
}

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1WA24CS218
Enter the number of processes: 5
Enter the number of resources: 3
Enter the allocation matrix:
Process 0: 0 1 0
Process 1: 2 0 0
Process 2: 3 0 3
Process 3: 2 1 1
Process 4: 0 0 2
Enter the request matrix:
Process 0: 0 0 0
Process 1: 2 0 2
Process 2: 0 0 0
Process 3: 1 0 0
Process 4: 0 0 2
Enter the available resources: 0 0 0

System is in safe state.
Safe Sequence is: P0 P2 P3 P4 P1

Process returned 0 (0x0)   execution time : 36.712 s
Press any key to continue.
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